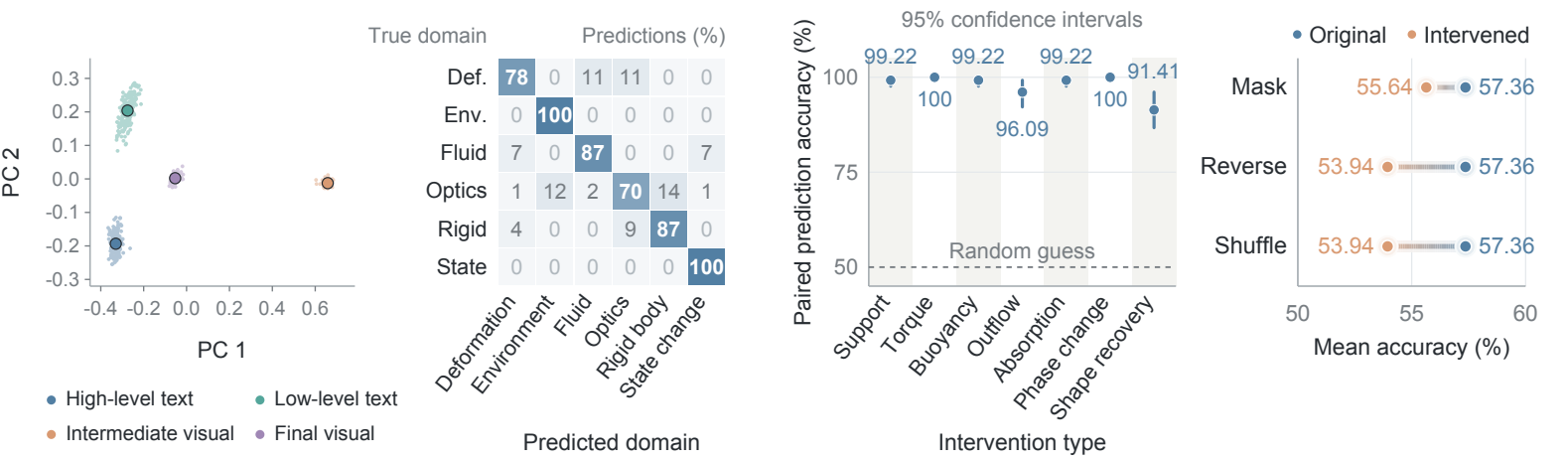


(a) Token-role structure **(b)** Physical-domain linear probe **(c)** Physical-intervention prediction **(d)** Visual-token perturbation